NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) with end-to-end robotics-software experience across navigation, SLAM, perception, and motion planning. Implemented the full self-driving stack in **C++ from scratch** — LiDAR perception (BEV YOLOv4 + EKF sensor fusion), localization (ICP / SVD, **0.299 m pose error / 4× baseline**, "outstanding" reviewer feedback), motion planning (FSM + cubic spiral trajectories), PID control — on Waymo Open Dataset + CARLA. Built a physical autonomous car as bachelor's capstone (Raspberry Pi + Arduino + OpenCV + PID) and trained visuomotor robot policies in MuJoCo (PyTorch + Diffusion Policy).

TECHNICAL SKILLS

C++ & Robotics Stack: C++ | CMake | Eigen | OpenCV | PCL (Point Cloud Library) | CARLA Simulator | MuJoCo | Raspberry Pi + Arduino | I2C / Embedded | Real-Time Systems | Python | Linux | Git

Perception, SLAM & Planning: LiDAR Point Cloud Processing | Bird's-Eye View Projection | 3D Object Detection | Extended Kalman Filter | Multi-Object Tracking | ICP / SVD (from scratch) | NDT | Scan Matching | YOLOv4 | SSD ResNet18 / 50 | Waymo Open Dataset | Behavior FSM | Cubic Spiral Trajectories | PID Control

ML on Robots & Tools: PyTorch | TensorFlow | Diffusion Policy | Visuomotor Policy Learning | LangGraph | Multi-provider LLM Routing | FastAPI | Docker | GitHub Actions CI/CD

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0) — **Expected May 2026**
Arizona State University — **Tempe, AZ**

Coursework: Perception in Robotics · Reinforcement Learning in Robotics · Advanced Computer Architecture · Algorithm/Hardware Co-Design · Foundations of Algorithms

Bachelor of Engineering — Electrical and Electronics — **August 2022**
BMS College of Engineering — **Bengaluru, India**

PROJECT EXPERIENCE

End-to-End Self-Driving Stack — C++ from Scratch, Waymo Open Dataset, CARLA — **Feb 2023**

- **Perception & Sensor Fusion:** Detected vehicles in 3D LiDAR point clouds via bird's-eye view projection using YOLOv4 (Darknet) and FPN ResNet18 on the Waymo Open Dataset; implemented an Extended Kalman Filter from scratch fusing LiDAR and camera measurements — full predict/update cycle, Mahalanobis distance data association, and track lifecycle (initialization, confirmation, deletion); validated via RMSE across multi-frame sequences.
- **SLAM / Localization:** Implemented ICP from scratch using SVD decomposition (KdTree nearest-neighbor association, centroid alignment, iterative convergence) and benchmarked against NDT and PCL-based ICP; achieved **0.299 m pose error over 172 m traveled** (requirement <1.2 m) — **4× better than baseline**, earning "outstanding" reviewer feedback.
- **Navigation, Motion Planning & Control:** Built an FSM behavior planner (lane follow, nudge, lane change, intersection stop) with a motion planner generating cubic spiral trajectories evaluated by collision and lane-deviation cost functions; implemented PID controllers from scratch for steering (cross-track error) and throttle in CARLA.

Autonomous Car — Real-Robot Deployment, Bachelor's Capstone (Raspberry Pi + Arduino) — **Jan – Jun 2022**

- Built and deployed a full physical autonomous car — OpenCV lane detection driving a PID steering controller, cascade classifiers for real-time traffic sign and obstacle detection (stop, U-turn, avoidance), Bluetooth remote control for manual data collection; Arduino motor driver controlled via I2C from Raspberry Pi, Python perception feeding C++ control logic on the robot.

Diffusion Policy — Visuomotor Policy Learning in Simulation (MuJoCo + PyTorch) — **Apr 2025**

- Adapted the Columbia Diffusion Policy architecture (RSS 2023) for learning robot control policies from demonstrations — configured MuJoCo simulation environments, ran end-to-end PyTorch training and rollout evaluation against simulated manipulation tasks; team project in CSE598 Perception in Robotics.

SiliconCrew — Autonomous LLM Agent for Code Generation & Verification — **Nov 2025 – Present**

- Built a LangGraph ReAct agent (21 tools) with multi-provider LLM routing (Gemini/OpenAI/Anthropic), autonomous self-correction loop (failure → diagnose → patch → re-verify with zero human input), and a full MCP server (stdio/SSE/HTTP); achieved 37/92 first-pass and 5/5 on structured code generation benchmarks.

Trezzit — Production AI SaaS (trezzit.com) — **Feb 2025 – Present**

- Sole-built and shipped a production AI SaaS serving **600+ users** — two-pass LLM receipt pipeline with provider-abstracted registry across Gemini, OpenAI, Claude, and Grok; GitHub Actions CI with 995 automated tests; live at trezzit.com.

WORK EXPERIENCE

Systems Engineer — **Oct 2022 – Apr 2024**
NCR GOLD — **Bengaluru, India**

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, a healthcare AI assistant with a custom RAG pipeline for insurance Q&A and automated dispute-email generation.